



**NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY**

Faculty of Computing and Informatics

Name: Tupopila Kadhila

Student Number: 223084433

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A Theory of Multilevel Interactive Equilibrium in NeuroAI

By: Zhe Sage Chen, Quanyan Zhu



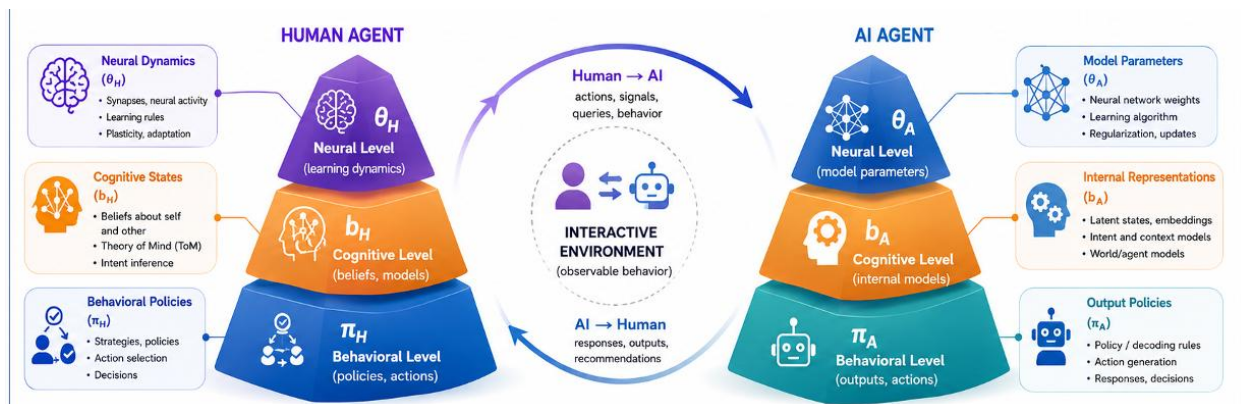
1. Summary of Paper

The Problem Being Addressed

One of the big gaps the authors point out is that our current theories of intelligence whether we're talking about humans or AI tend to be a bit too optimistic. Classical game theory often assumes that everyone is perfectly rational and has infinite brain power to make decisions. In reality, our brains are limited by things like "partial observability" (we can't read minds), finite computation, and a whole lot of uncertainty. The most existing models only look at **behavior** "what we actually *do*" but they ignore the "under-the-hood" stuff like our internal beliefs or what's happening at the neural level.

The Main Contribution of the Work

The paper introduces a framework called **Multilevel Interactive Equilibrium (MIE)**. Think of it as a way to broaden the classic Nash equilibrium. Instead of just looking at stable behaviors, MIE says that true stability in an interaction (like between a person and an AI) happens across three layers: the **neural level** (the learning substrate), the **cognitive level** (our internal beliefs and mental models), and the **behavioral level** (our actual strategies). It's a unifying theory that works whether you're looking at two humans, two AI agents, or a hybrid human-AI system.



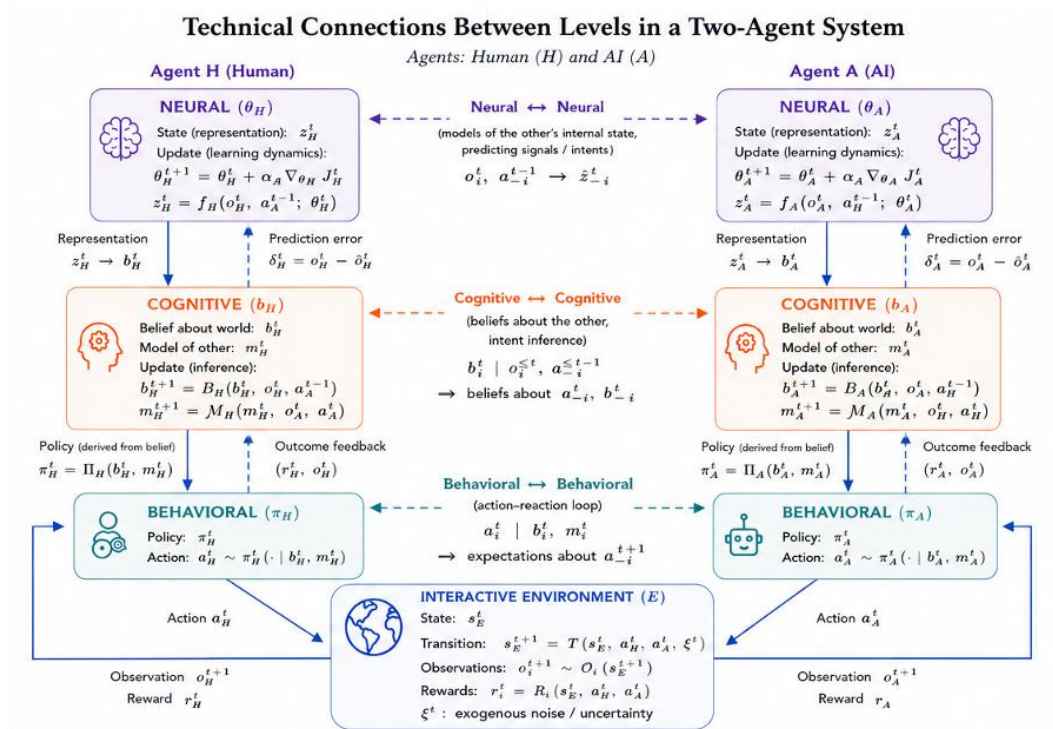
*The Multilevel Interactive Equilibrium (MIE) framework. This pyramid illustrates how agents whether human or AI are composed of three interconnected layers: **Neural** (learning), **Cognitive** (beliefs), and*

Behavioral (actions). Stability occurs when all three levels across both agents align through their interaction in a shared environment

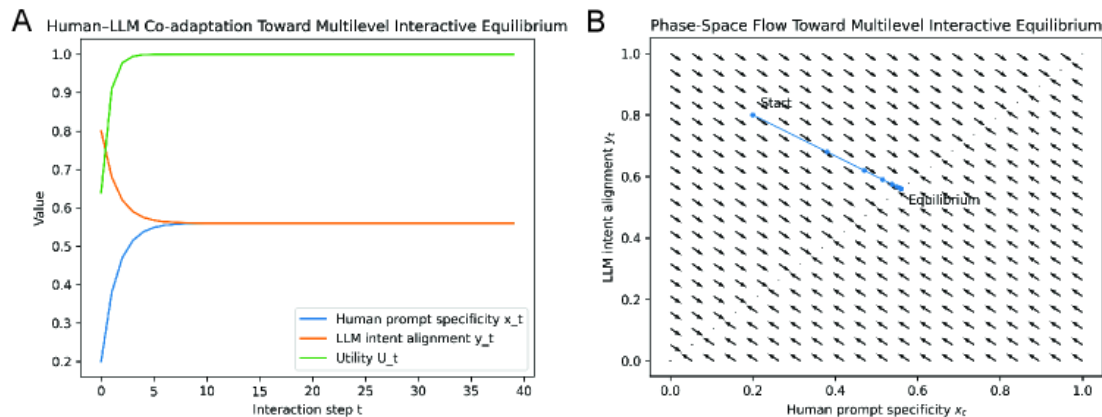
The Experimental/Theoretical Results

The authors basically turn this concept into a mathematical "global dynamical system". They define equilibrium as a fixed point where all three levels neural, cognitive, and behavioral stabilize at the same time. They didn't just leave it as math, though; they applied the theory to real-world scenarios like:

- **Self-driving cars:** How a human driver and an autonomous vehicle can reach a stable "understanding" on the road.
- **Brain-Machine Interfaces (BMIs):** How a brain and a prosthetic device co-adapt to work together.
- **Human-LLM interaction:** They even ran a simulation showing how a user's prompt specificity and an AI's alignment converge into a stable, efficient conversation.



The mathematical 'wiring' of the system. This diagram shows how internal updates (like neural prediction errors and cognitive belief shifts) drive an agent's policy, while external feedback from the environment couples the two agents together in a continuous loop of action and reaction



A simulation of Human-LLM co-adaptation. **Graph A** shows how task utility (success) increases as the human's prompts and the AI's understanding converge. **Graph B** tracks this journey in a 'phase-space,' showing how different starting points naturally flow toward a stable state of mutual understanding, or MIE

2. Related Work

The authors aren't starting from scratch; they're standing on the shoulders of some pretty heavy hitters. They build directly on John von Neumann and Oskar Morgenstern's classical game theory and the Nash equilibrium. They also pull in ideas from Reinforcement Learning (like Littman's Markov games) and Decision Neuroscience to explain how the brain handles social variables like prediction errors. You can also see the influence of Active Inference (from Parr and Friston) and newer research into "neuronic Nash equilibrium" which looks at neural synchronization in EEG data.

3. Limitations of the Paper

As ambitious as this framework is, the authors are pretty honest about where it still falls short:

- **Mathematical Proofs:** While the Nash equilibrium has well-established rules for when it exists, we don't yet have those same proofs for MIE. It's still unclear how often these complex multilevel systems actually reach a unique stable state.

- **Measuring the "Hidden" Stuff:** A major hurdle is that neural weights and cognitive beliefs are **latent variables** you can't see them directly. Trying to estimate these from real-world data requires massive, complicated models that are hard to fit.
- **Timing Issues:** Neural learning, belief updates, and behavioral reactions happen at different speeds (some fast, some slow). The paper notes that we still need better tools to understand how equilibrium emerges when these "clocks" are out of sync.
- **The "Human Factor":** The model might be missing things like **cultural norms, group pressure, or impulsive drives** that don't always fit into a clean "cognitive" or "neural" box.